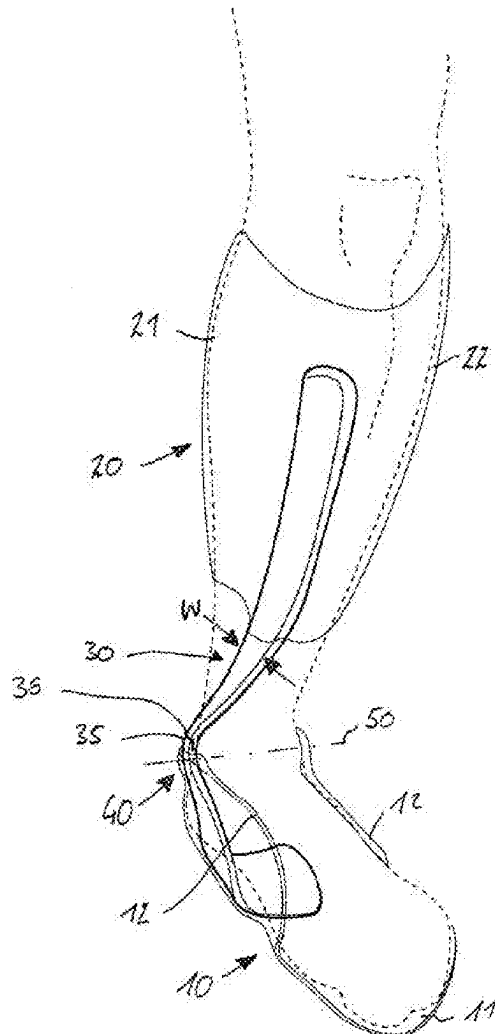




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Alexander SCHMITT, Kassel (DE)(21) Appl. No.: **19/210,476**(22) Filed: **May 16, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/539,596, filed on
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(2013.01)(57) **ABSTRACT**

An ankle/foot orthosis with a foot part, which has a sole for receiving a foot, and with a shin part which, when fitted in place, bears on the frontal aspect of a shin and is connected to the foot part via a medially extending spring, wherein the spring, behind an ankle area, forms a hinge area that allows a movement of the shin part relative to the foot part in the anterior-posterior direction.



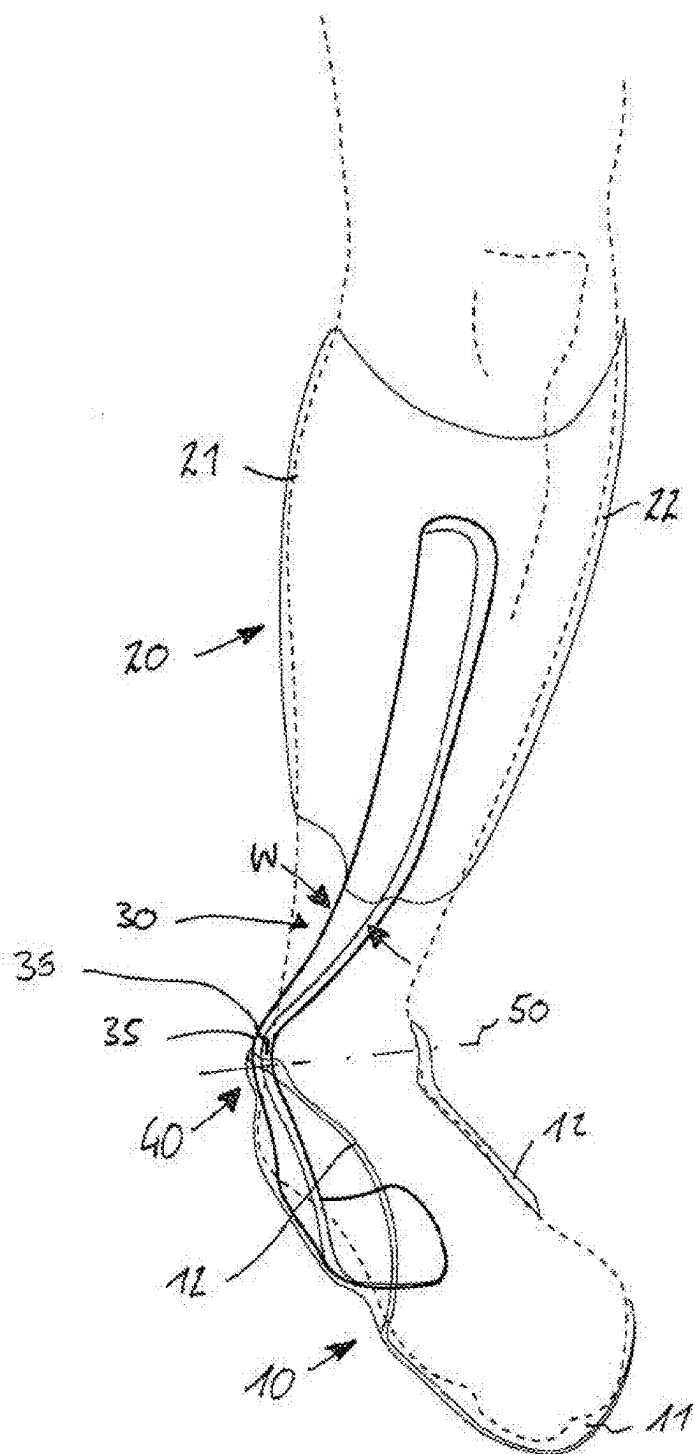
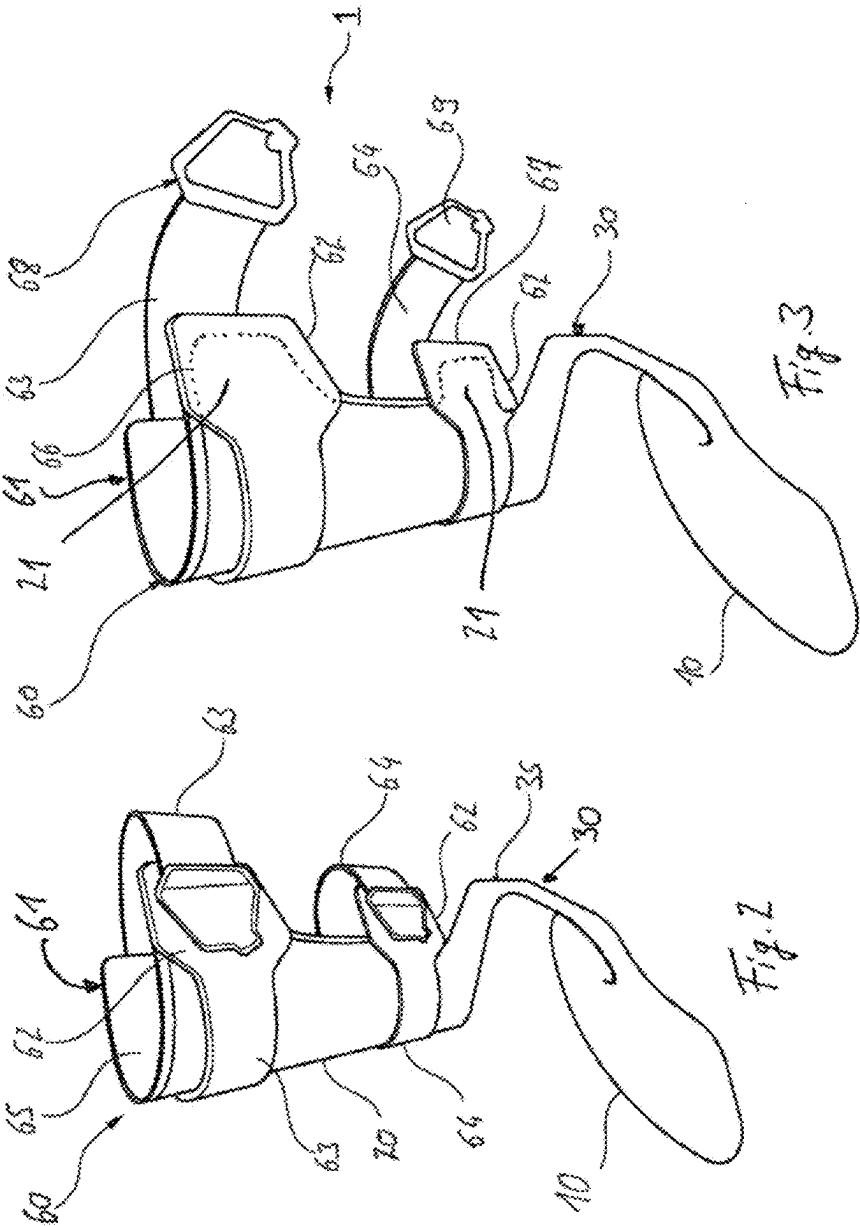


Fig. 1



ANKLE/FOOT ORTHOSIS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation application of U.S. patent application Ser. No. 18/539,596, filed 14 Dec. 2023, and entitled ANKLE/FOOT ORTHOSIS, which is a continuation of U.S. patent application Ser. No. 18/074,258, filed 2 Dec. 2022, and entitled ANKLE/FOOT ORTHOSIS, now U.S. Pat. No. 11,850,177, which is a continuation application of U.S. patent application Ser. No. 16/926,595, filed 10 Jul. 2020, now U.S. Pat. No. 11,517,463, and entitled ANKLE/FOOT ORTHOSIS, which is a continuation application of U.S. patent application Ser. No. 14/406,835, filed 10 Dec. 2014, now U.S. Pat. No. 10,729,574, and entitled ANKLE/FOOT ORTHOSIS, which is a U.S. National Entry Application from PCT International Patent Application No. PCT/EP2013/001663, filed 6 Jun. 2013, and entitled ANKLE/FOOT ORTHOSIS, which claims the benefit of German Patent Application No. 102012011466.4, filed 12 Jun. 2012, the disclosures of which are incorporated, in their entireties, by this reference.

TECHNICAL FIELD

[0002] The invention relates to an ankle/foot orthosis with a foot part, which has a sole for receiving a foot, and with a shin part which, when fitted in place, bears on the frontal aspect of a shin and is connected to the foot part via a spring.

BACKGROUND

[0003] If the ankle joint lacks stability and the muscle strength is inadequate, it may be useful and necessary to use an ankle/foot orthosis. The ankle/foot orthosis keeps the foot in a relationship to the shin and prevents unwanted dorsal flexion, and also plantar flexion when the foot is unloaded. During walking, the orthosis assists the patient by maintaining a dorsally flexed position at heel strike and by allowing pushing-off in the terminal stance phase. Unwanted pronation and supination of the foot are likewise avoided.

[0004] DE 603 15 698 T2 relates to an ankle/foot orthosis with a structural frame which has a foot plate, an inner part and an ankle part and which splits into two frontal support elements, which are arranged medially and laterally on the frontal aspect of the shin. The ankle/foot orthosis can be fitted on the patient using a fastening device in the form of a velcro fastener.

SUMMARY

[0005] The object of the present invention is to make available an ankle/foot orthosis which permits a high level of functionality, allows simple insertion into the shoe and is easy to fit on the patient.

[0006] According to the invention, this object is achieved by an ankle/foot orthosis having the features of the main claim. Advantageous embodiments and developments of the invention are disclosed in the dependent claims, in the description, and in the figures.

[0007] In the ankle/foot orthosis according to the invention, with a foot part, which has a sole for receiving a foot, and with a shin part which, when fitted in place, bears on the frontal aspect of a shin and is connected to the foot part via a spring, provision is made that the spring, behind an ankle area, forms a hinge area that allows a movement of the shin

part relative to the foot part in the anterior-posterior direction. The course of the spring behind an ankle area, i.e., the area where the natural axis of the ankle joint lies, ensures that no spring material presses against the ankle or the shin.

[0008] In a development of the invention, provision is made that the spring is arranged extending medially. The medial course of the spring prevents the spring from catching on objects during walking, in addition to which the medial course of the spring ensures an unobtrusive positioning of the ankle/foot orthosis.

[0009] Moreover, the spring can in this way be connected to the foot part in the area of the plantar arch, as a result of which the course of the spring in the transition area can be more flowing than in the lateral sole area.

[0010] The hinge area of the spring is designed to be torsionally stable about an axis that runs from proximal to distal, i.e., corresponds substantially to a vertical direction. It is thus possible that the orientation of the foot relative to the shin is maintained during walking and there is only a spring movement, on account of the hinge area, in the sagittal plane, i.e., about an axis extending transversely with respect to the direction of walking.

[0011] The hinge area is advantageously designed to be flexurally stiff in the medial-lateral direction, in order to counteract a pronation movement or supination movement of the foot or of the foot part.

[0012] In one embodiment of the invention, provision is made that the spring extends medially inward from the foot part and obliquely upward in the dorsal direction, such that it protrudes behind the ankle area. From the ankle area, it then extends obliquely upward in the ventral direction. The course of the spring is therefore such that a lower spring portion extends obliquely rearward at an articulation of the spring in the area of the plantar arch, wherein the front edge of the spring lies behind the axis of the joint. From the farthest posterior position of the spring, an upper portion of the spring then extends obliquely forward in order to support the shin.

[0013] To form the hinge area, provision is made that the spring has a tapering in the anterior-posterior direction, such that, on account of the narrowed configuration of the spring in the area with reduced material, a defined zone is provided that facilitates yielding in the anterior-posterior direction.

[0014] In order to make dorsal flexion and plantar flexion easier, the spring can have a lower section modulus in the anterior-posterior direction than in the medial-lateral direction. This lower section modulus can be achieved by a specific geometric configuration of the spring; different section moduli in the respective orientations can be provided in particular in the hinge area.

[0015] To permit pivoting in the anterior-posterior direction and at the same time to avoid pivoting in the medial-lateral direction, so that the foot remains on the foot part in the adopted orientation, provision can also be made that the spring has, in the hinge area, a material structure that has a greater elasticity in the anterior-posterior direction than in the medial-lateral direction. This can be achieved, for example, by a specific choice of fiber-reinforced materials for the spring. For example, carbon can be used as fiber material in the medial-lateral direction, carbon having less elasticity than, for example, glass fiber, aramid fiber or polyethylene fiber, which are responsible for the stiffness or elasticity in the anterior-posterior direction. The different elasticities can therefore be achieved not only by geometric

variations but also by the suitable orientations of the fiber reinforcements within the spring.

[0016] The shin part can have a gutter shape or cup shape and is flexurally stiff in the medial plane, such that a stable contact of the shin on the shin part is permitted. By virtue of the flexurally stiff configuration in the medial plane, it is possible to achieve a good transfer of force from the foot part to the shin part via the spring, without this leading to punctiform loads in the shin area.

[0017] The spring can be curved in the medial direction with respect to a sagittal plane, in order to extend medially inward from the foot part, so as not to touch the ankle protruding medially to the contralateral side but instead to run around it both medially and also in the posterior direction.

[0018] The shin part can have a padding to make the orthosis easier to wear. The padding can be secured on the shin part via pockets on projections. It is likewise possible that the pockets bear medially and laterally on the entire shin part, such that the shin part is received, at least at the edges, completely in the pockets. To secure the orthosis to the leg and also the padding to the shin part, fastening elements can be provided which are guided from the lateral side to the medial side in order to make fitting and securing easier. The spring can be designed as an arc and, as a separate component, can be connected to a separate foot part and a separate shin part. It is also possible in principle that the ankle/foot orthosis is in one piece, such that the spring is an integral component both of the foot part and also of the shin part. The spring itself can be designed as a separate part and can be connected to the respective adjoining components via plug connections, if appropriate by form-fit securing, or an adhesive connection.

[0019] Furthermore, provision can be made that the spring in the ankle area has the least stiffness against bending about an ankle joint axis, thereby permitting mobility in the anterior-posterior direction. During walking or standing, the point of introduction of force into the foot or into the foot part changes according to the phase of movement or the shifting of weight. The forces acting on the ankle joint cause moments about the respective joint axis, wherein, at a constant force, the active moment increases with the distance from the joint axis. On account of the geometric relationships, the greatest moment is therefore applied around the axis of the ankle joint, since the point of introduction of force is at a maximum distance from the axis of the ankle joint or the hinge area, whereas, on account of the geometric conditions, only a small moment is applied to the ankle area, and therefore also to the hinge area, about the longitudinal axis of the foot. This results in a relative stiff configuration against a supination movement or pronation movement, since, on account of the short distance from the hinge area, a smaller moment is active, whereas a relatively soft and elastic configuration is present for a plantar flexion and dorsal flexion, since greater yielding is afforded on account of the greater active moment around the hinge area.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Illustrative embodiments of the invention are explained in more detail below with reference to the attached figures, in which:

[0021] FIG. 1 shows an ankle/foot orthosis in a schematic set-up;

[0022] FIG. 2 shows an ankle/foot orthosis with padding; and

[0023] FIG. 3 shows an ankle/foot orthosis with open fastening elements.

[0024] FIG. 1 shows an ankle/foot orthosis in a schematic view, with a foot part **10** and a shin part **20**, which are connected to each other via a spring. The foot part **10** has a sole **11** on which it is possible to position a foot indicated by the broken line. In the metatarsal region, elevations are provided medially and laterally on the sole **11**, in order to support the foot both in the medial and also in the lateral direction. The elevations **12** can also enclose the heel. The foot part **10** is advantageously in one piece and is made of a dimensionally stable, slightly elastic material, which allows it to perform a heel-to-toe movement of the foot during walking. The stability is chosen such that sufficient support of the foot is ensured.

[0025] The shin part **20** has a cup shape or gutter shape and follows the shape of the natural leg. The shin part **20** can be flexurally stiff in the medial plane, such that there is firm contact along the longitudinal extent of the shin. In order to permit variations in volume during wear or also to permit adaptation to different users, the shin part **20** can be elastic in the medial and/or lateral direction at its end areas, such that the gutter-shaped shin part **20** can widen or, by suitable fastening elements, can be moved toward the lower leg. It is likewise possible that the shin part **20** is elastic only in one direction, i.e., either medially or laterally, since it is sometimes advantageous for one contact to be stiff while the other contact yields elastically. For example, if the medial contact is stiff, a rotary movement of the leg can be prevented.

[0026] Both components, i.e., the shin part **20** and also the foot part **10**, can be produced from fiber-reinforced plastics and are advantageously made in one piece. In an alternative embodiment, a multi-part configuration both of the foot part **10** and also of the shin part **20** is possible.

[0027] In the illustrative embodiment shown, the foot part **10** is connected to the shin part **20** via a medially arranged spring **30**. Foot part **10**, shin part **20** and spring **30** are designed as separate components and fastened to one another, for example via a plug connection, which is subsequently fixed, for example by a form-fit element or an adhesive bond. Alternatively, the foot part **10**, the shin part **20** and the spring **30** are permanently interconnected, for example adhesively bonded, welded, baked or laminated onto to one another. It is also possible in principle to produce both the foot part **10** and also the spring **30** and the shin part **20** in one piece from fiber-reinforced plastics or another material.

[0028] The spring **30** extends medially from the underside of the sole **11** of the foot part **10** and proximally across the plantar arch in the direction of the natural ankle, wherein the course from the sole **11** to an ankle area **40** runs obliquely in the posterior and proximal direction, i.e., obliquely rearward and upward, wherein the front edge of the spring **30** runs to a position behind the axis **50** of the natural ankle joint. From there, the spring **30** runs obliquely in the anterior and proximal direction, i.e., obliquely forward and upward, until it extends proximally, i.e., upward, substantially parallel to the shin. The shin part **20** is fastened to the proximal, upper end of the spring **30**.

[0029] In the ankle area **40** of the ankle/foot orthosis, the spring **30** forms a hinge area **35**, which lies approximately at the height of the axis **50** of the ankle joint. In this hinge

area 35, a flexion of the spring 30 is permitted substantially about the axis 50 of the natural joint, such that the shin part 20 can be moved elastically relative to the foot part 10. To form the hinge area 35, provision is made that the cross section changes from distal to proximal along the course of the spring 30 and has, in the hinge area 35, the lowest section modulus against a movement in the anterior-posterior direction, i.e., in the sagittal plane. For this purpose, the spring 30 in the hinge area 35 is provided, approximately at the height of the axis 50 of the natural ankle joint, with a tapering 36, such that the lowest section modulus is present in this area. The thickness of the spring 30, i.e., the extent in the medial-lateral direction, can be made substantially constant.

[0030] From the sole 11, the spring 30 firstly extends medially inward, in order then to extend in an arc in the lateral direction behind the ankle. After the hinge area 35, the spring 30 extends laterally in an arc shape until it merges into the shin part 30. The spring 30 thus extends in two arcs, wherein the greatest medial curvature lies below the hinge area 35 or in the hinge area 35, while the maximum posterior curvature lies behind the ankle area 40 of the ankle joint.

[0031] FIG. 2 shows a variant of the ankle/foot orthosis, with a shin part 20, a fastening means 60, and a foot part 10, which is connected to the shin part 20 via the spring 30. The spring 30 in this case permits only a slight, defined movement of the shin part 20 relative to the foot part 10. The foot part 10 is designed as a flat sole plate, which substantially follows the outer contour of a foot or of a shoe. The illustrative embodiment shown is in one piece and made from a fiber-reinforced plastic.

[0032] In the fitted state, the shin part 20 bears on the shin of the orthosis user and at least partially encloses the shin medially and laterally. The contour of the shin part 20 is substantially gutter-shaped or cup-shaped and can have recesses or apertures to permit improved ventilation of the shin area. A fastening means 60 is arranged on the inner side of the shin part 20 facing toward the orthosis user in the fitted state, said fastening means 60 at the same time being provided with a padding 65, such that the hard, cup-like structure of the shin part 20 does not lie directly on the lower leg.

[0033] Receiving regions 21 in the form of projections or tongues, which are shown in FIG. 3, are arranged both medially and also laterally on the shin part 20. The receiving regions 21 or projections are formed at the proximal end and distal end of the shin part 20 and, when the orthosis is in the fitted state on the orthosis user, face toward the rear. The receiving regions 21 are referred to below as projections, which term is also meant to include a configuration in which structures adjoining the shin part 20 are suitable for being received in pockets in order to bring about a form-fit securing of the padding 65.

[0034] On the fastening means 60, pockets 61, 62 are formed into which the projections or receiving regions 21 are pushed. By pushing the projections or receiving regions 21 into the pockets 61, 62, the fastening means 60 is secured on the shin part 20. Provision is made here that the lateral pockets 61 are deeper than the medial pockets 62, and provision is likewise made that the lateral projections or receiving regions 21 are longer than the medial projections or receiving regions 21, wherein the depth of the pockets 61, 62 corresponds to the lengths of the projections 21. The illustrative embodiment provides two pockets 61, 62 and two projections 21 on the medial and lateral sides, respec-

tively, which are offset relative to each other in the longitudinal extent, such that two proximal and two distal pairs of pockets 61, 62 and projections or receiving regions 21 result. Alternatively, it is also possible for just one pocket to be provided medially and laterally and to be secured on a corresponding receiving region 21.

[0035] Fastening elements 63, 64 are provided on the fastening means 60, both at the proximal edge and also at the distal edge of the shin part 20, via which fastening elements 63, 64 the orthosis is secured on the limb. The fastening elements 63, 64, in the form of straps or bands, are secured, for example welded, adhesively bonded or sewn, on the medial side on the fastening means 60 to form the medial pockets 62. From the medial pocket 22, the connection element 63, 64 designed as a strap is guided externally along the outer side of the shin part 20 and is secured externally on the lateral pocket 61, for example by a velcro fastener. In the case of an elastic configuration of the strap-like fastening element 63, 64, the medial and lateral pockets 61, 62 are pretensioned toward each other in the circumferential direction, such that the forwardly open pockets 61, 62 cannot slide down from the rearwardly pointing projections 21. To further secure the orthosis, the respective connection element 63, 64 is then guided rearward around the limb (not shown) from the lateral side to the medial side and is secured there on a fixing means 66, 67. The connection element 63, 64 is thus guided once around the orthosis and the limb, wherein the free end is secured on the outer side of the medial pocket 62. At the free end of the fastening element 63, 64, locking elements 68, 69 are secured, for example in the form of detachable velcro elements, which can be secured on the outer side of the fastening element 63, 64 or on fleeced areas of the fixing means 66, 67 externally on the medial pockets 62. The securing on the fastening elements 63, 64 and on the fixing means 66, 67 is reversible. The fastening elements 63, 64 can be flexible and nonelastic, flexible and elastic, or elastic only in parts, and it is likewise possible that the connection element 63, 64 is secured reversibly on the fastening means 60, for example by velcro fasteners.

[0036] In the fitted state as shown in FIG. 2, the connection element 63, 64 bridges the gap in the cup-like shin part 20 from lateral to medial. The shin part 20 can permit a deformation of the projections 21 toward each other, such that an adjustment to the dimensions of the limb on which the orthosis is to be fitted can take place. The adjustment can be made via the circumferential force applied by the fastening elements 63, 64.

[0037] The embodiment shown is especially advantageous for patients who are paralyzed on one side or weakened on one side. When the connection element 63 is guided from the medial side to the lateral side, stretched and secured with tensioning on a lateral pocket 61, the hand that is unaffected by the weakness can be used to safely secure the orthosis by then guiding the connection element 63, 64 behind the limb and locking it on the medial side.

[0038] The fastening means 60 can also be designed as a one-part element or as a multi-part, permanently interconnected element that has a cushioning function. The medial and lateral projections 21 form self-adaptive side wings which adapt automatically to the shape of the limb under the circumferential force that is applied by the connection element 63, 64.

[0039] With Y-shaped velcro tapes 68, 69 as hook areas, it is possible to easily shorten the fastening elements 63, 64 designed as straps, by removing the Y-shaped Velcro tapes from the fleeced fastening elements 63, 64, shortening them to the desired length and then reapplying them.

[0040] Particularly in the case of stroke patients, it must be noted that they are able to fit the orthosis using only the unaffected side of the body. By virtue of the fastening elements 63, 64 being arranged laterally and being guided from lateral to medial in order to secure the orthosis, it is possible for the orthosis user, in a seated position, to grasp the connection element 63, 64 laterally, using his or her unaffected side, and to simply pull it in the medial direction and secure it there on the hook and loop areas 66, 67 provided there.

1. An ankle/foot orthosis, comprising:
 - a foot part, which has a sole to receive a foot, and with a shin part which, when fitted in place, bears on the frontal aspect of a shin and is connected to the foot part via a spring, wherein the spring, behind an ankle area, forms a hinge area that allows a movement of the shin part relative to the foot part in a anterior-posterior direction.
2. The ankle/foot orthosis as claimed in claim 1, wherein the spring extends medially.
3. The ankle/foot orthosis as claimed in claim 1, wherein the hinge area is designed to be torsionally stable about a proximal-distal axis.
4. The ankle/foot orthosis as claimed in claim 1, wherein the hinge area is designed to be flexurally stiff in a medial-lateral direction.
5. The ankle/foot orthosis as claimed in claim 1, wherein the spring extends obliquely upward in a dorsal direction from the foot part and extends obliquely upward in a ventral direction behind the ankle area.
6. The ankle/foot orthosis as claimed in claim 1, wherein the spring has a tapering in the anterior-posterior direction.
7. The ankle/foot orthosis as claimed in claim 1, wherein the spring has a lower section modulus in the anterior-posterior direction than in a medial-lateral direction.
8. The ankle/foot orthosis as claimed in claim 1, wherein the spring has, in the hinge area, a material structure that has a greater elasticity in the anterior-posterior direction than in a medial-lateral direction.

9. The ankle/foot orthosis as claimed in claim 1, wherein the shin part has a gutter shape or cup shape and is flexurally stiff in a medial plane.

10. The ankle/foot orthosis as claimed in claim 1, wherein the shin part is elastic in at least one of a medial direction and a lateral direction.

11. The ankle/foot orthosis as claimed in claim 1, wherein the spring is curved in a medial direction with respect to a sagittal plane.

12. The ankle/foot orthosis as claimed in claim 1, wherein the shin part has a padding.

13. The ankle/foot orthosis as claimed in claim 12, wherein the padding is secured via pockets on projections or receiving regions.

14. The ankle/foot orthosis as claimed in claim 11, wherein the padding has fastening elements, which are guided from a lateral side to a medial side.

15. The ankle/foot orthosis as claimed in claim 1, wherein the spring is designed as an arc.

16. The ankle/foot orthosis as claimed in claim 1, wherein the spring in the ankle area has the least stiffness against bending about an ankle-joint axis.

17. An ankle/foot orthosis, comprising:

- a foot part, which has a sole to receive a foot;
- a shin part which, when fitted in place, bears on the frontal aspect of a shin;
- a spring configured to connect the shin part to the foot part;
- an ankle area;
- a hinge area formed by the spring behind the ankle area, the hinge area allowing movement of the shin part relative to the foot part in a anterior-posterior direction.

18. The ankle/foot orthosis as claimed in claim 17, wherein the spring extends medially.

19. The ankle/foot orthosis as claimed in claim 17, wherein the hinge area is designed to be torsionally stable about a proximal-distal axis.

20. The ankle/foot orthosis as claimed in claim 17, wherein the hinge area is designed to be flexurally stiff in a medial-lateral direction.

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